

R1 Mock 3 — Project-first pressure

20 minutes · Ojas Srivastava · Intern – Technology

Morning-of mock. No notes. They interrupt.

From reports: Shivam Kumar (ex-GEP intern, campus SDE): questions from intro, code, resume. AmbitionBox: “Explain key projects.” GFG intern R1: project difficulties + string DSA. Sarthak: login/auth follow-up pattern.

Interviewer tone: opens on projects, interrupts, squeezes OOP later.

How to use. Set a 20-minute timer. Cover the green boxes. Lead with LogiFlow without waiting for an OOP prompt.

Do NOT volunteer: location, IFFCO (unless they open resume), MCP names, AirHelp, “I led the whole team.” Do not say ranking by “type of goods.”

Locked intro — say verbatim, then stop:

Good afternoon sir, ma’am. I am Ojas Srivastava. Pre-final B.Tech AI, SVNIT Surat, CGPA 9.20. Coursework: OOP, DBMS, OS, CN, DAA. Google Solution Challenge 2026 Global rank 106 — LogiFlow, supply-chain routing. Vibe2Ship 2026 Global Top 20 — Community Hero, civic issue reporting PWA. Coding workflow: Cursor, MCP and plugins; spec and review every merge. Chairperson Nexus SVNIT, CSE+AI cell, manage and mentor 500+ students. I own the outcome. GEP works on AI-powered procurement and supply chain — operational data to a decision, owned till it ships.

1. Clock

0:00–2:00	Intro	Locked intro. Pick LogiFlow. They interrupt.
2:00–10:00	LogiFlow	Ranking, pipelines, tests, Community Hero auth.
10:00–14:00	OOP	Compressed — composition, overloading vs overriding.
14:00–18:00	DSA	Remove repeating characters. Bonus: seconds between times.
18:00–20:00	Close	Difficulty faced. They cut you off.

2. Script

0:00 — Intro + project pick

Interviewer. *Tell me about yourself. Pick one project — I’ll interrupt.*

You say. Intro, then: “I’d start with LogiFlow — closest to GEP’s supply chain work.”

2:00 — LogiFlow first

Interviewer. *What does LogiFlow rank on? How do you define delay risk?*

You say. Time, cost, delay risk. Delay from historical delay patterns on corridor train-day records — operational data, not guessed.

Interviewer. *Railway vs comparator vs hybrid — difference?*

You say. Railway = rail-specific path. Comparator = compares modes/options. Hybrid = combines sources. All feed same ranker interface.

Interviewer. *pytest 100/100 — what kind of tests?*

You say. Business-rule regression — broken pricing/routing fails in CI before release. Not just smoke tests.

Interviewer. *Community Hero on resume — how did you do auth?*

You say. Firebase token on client; Express verifies on every protected route. Roles server-side. Never trust client for authorisation.

10:00 — OOP (compressed)

Interviewer. *How did you use OOP in LogiFlow?*

You say. Pipeline objects + ranker object. Each pipeline encapsulates fetch/transform. Ranker sees common interface.

Interviewer. *Inheritance vs composition — which did you prefer?*

You say. Composition — Ranker *has* pipelines, does not inherit. Easier to swap railway for hybrid at runtime.

Interviewer. *Overloading vs overriding?*

You say. Overloading = compile-time, different signatures. Overriding = runtime, subclass replaces parent.

14:00 — DSA

Interviewer. *Remove repeating characters, keep first-occurrence order.*

You say. One pass + seen set. “programming” → “progamin”. $O(n)$.

Interviewer. *Bonus if fast: seconds between 23:50:00 and 00:10:00?*

You say. Convert to seconds, subtract; if negative add 86400 → **1200 seconds**.

18:00 — Close

Interviewer. *What difficulty did you face?*

You say. Latency. First version recomputed every request. Cache-aside Redis, SQL as source of truth — 100–400 ms.

Interviewer. *We’re out of time. Thank you.*

3. Score this mock

Rate each area /10. Pass bar: **7+** average.

Intro	Picked LogiFlow without rambling? Stopped when interrupted?	_____
OOP	Composition over inheritance? Overload vs override correct?	_____
LogiFlow	Delay from records, not “type of goods”? 100/100 = business rules?	_____
DSA	One-pass + seen set? Midnight wrap = 1200s if asked?	_____
Close	Latency story complete in two sentences?	_____

Pass bar. Led with LogiFlow without waiting for an OOP prompt. String DSA one-pass with complexity. If this is the morning of: stop here. Do not cram more PDFs.